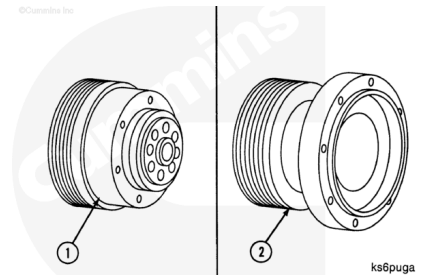


Trainings ▾

General Information

There are two pulley designs used on the QSK19 series engines. Pulley (1) is for engines that have one vibration damper. Pulley (2) is for engines that have two vibration dampers.

Note : The number of grooves in the pulley can be different than illustrated.

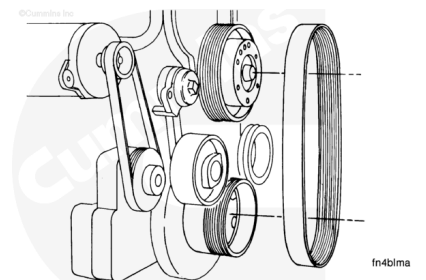


Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

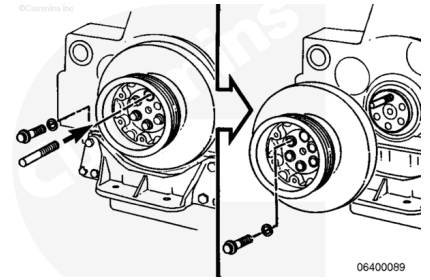
- Remove the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>).
- Remove the vibration damper and crankshaft pulley. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/20/20-001-052-tr.html>).



Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



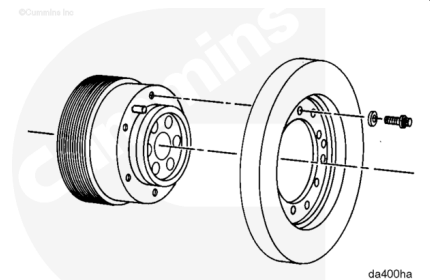
⚠ CAUTION ⚠

Do not pry or hammer on the vibration damper. Damage to the vibration damper can result.

Remove one capscrew and install a [3/4-16x5 in] guide stud.
Remove the remaining capscrews from the vibration damper.
Remove the guide stud.

Note : Engines that do not have a belt driven fan hub have an adapter instead of a pulley.

Remove the pulley from the vibration damper.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriated eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

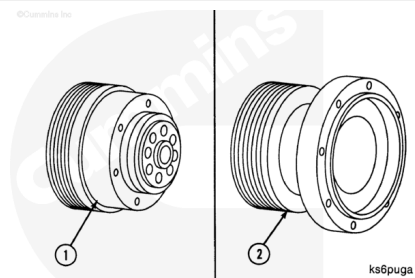
Use solvent or steam to clean the pulley and dry with compressed air.

Check all mounting pilots and surfaces for damage from fretting.

If it is **not** possible to remove the damage with a 240-grit abrasive cloth, the crankshaft pulley **must** be replaced.

⚠ CAUTION ⚠

The mounting pilots and surfaces control the center and spacing of the pulleys. Do not repair the part by machining. It will cause out-of-balance condition or a crankshaft failure.



Measure the outside diameter.

mm		in
134.78	MIN	5.3105
135.04	MAX	5.3115

Measure the inside diameter

mm		in
110.780	MIN	4.3610
110.806	MAX	4.3620

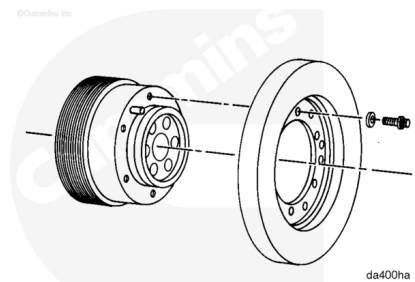
Install

Install the vibration damper on the crankshaft pulley or adapter (if equipped).

Install the six washers and capscrews.

Tighten the capscrews.

Torque Value: 140 n•m [105 ft-lb]

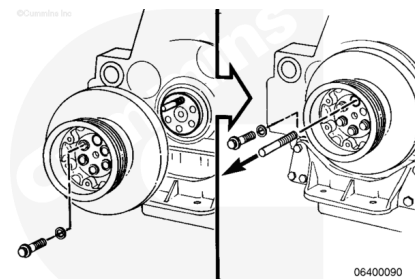


Install the guide stud in the crankshaft.

Install the vibration damper assembly. The pilot **must** be aligned with the crankshaft correctly.

Install the washers and capscrews.

Tighten the capscrews.

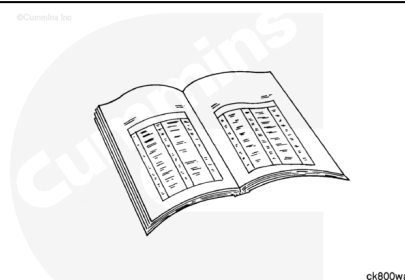


Torque Value:	Step 1	230 n.m [170 ft-lb]
	Step 2	445 n.m [330 ft-lb]

Clean and Inspect for Reuse

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the vibration damper and crankshaft pulley. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/20/20-001-052-tr.html>).
- Install the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>).

Operate the engine and check for noise.